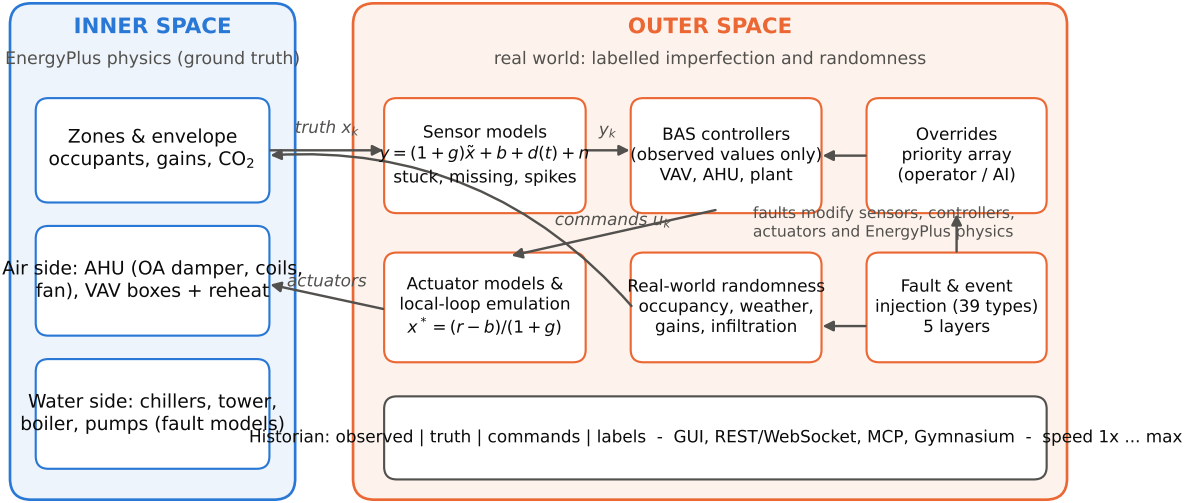
INNER SPACE
EnergyPlus physics (ground truth)
Zones & envelope
occupants, gains, CO2
Air side: AHU (OA damper, coils, fan), VAV boxes + reheat
Water side: chillers, tower, boiler, pumps (fault models)
truth x_k
actuators
OUTER SPACE
real world: labelled imperfection and randomness
Sensor models
$y = (1+g)\tilde{x} + b + d(t) + n$
stuck, missing, spikes
y_k
BAS controllers
(observed values only)
VAV, AHU, plant
Overrides
priority array
(operator / AI)
commands u_k
faults modify sensors, controllers, actuators and EnergyPlus physics
Actuator models & local-loop emulation
$x^* = (r-b)/(1+g)$
Real-world randomness
occupancy, weather, gains, infiltration
Fault & event injection (39 types)
5 layers
Historian: observed | truth | commands | labels - GUI, REST/WebSocket, MCP, Gymnasium - speed 1x ... max